

ABDUL RAB

@ abdulrab@live.com +92 (333) 7182996 Al Madina Colony, Near Board Office, Sukkur, Pakistan

in www.linkedin.com/in/abdulrab/

EDUCATION

Master of Science in Mathematics

 Jan 2020 – Dec 2022

 Sukkur IBA University

CGPA 3.48/4 and Percentage: 86.88

Bachelor of Science in Mathematics

 Aug 2015 – Jul 2019

 Sukkur IBA University

CGPA 2.56/4 and Percentage: 75.78

EXPERIENCE

Lecturer in Mathematics (BPS-17)

Government Islamia Science College, Sukkur

 Oct 2025 – Present

 Sukkur, Sindh, Pakistan

Teaching (B.S. Computer Science): Differential Equations; Probability and Statistics; Numerical Analysis; Calculus and Analytical Geometry.

Mathematics Teacher (Visiting Faculty)

Sukkur IBA University

 Oct 2019 – Present

 Sukkur, Sindh, Pakistan

Courses Taught:

- Fall 2025: Multivariate Calculus (Computer Science)
- Summer 2025: Linear Algebra (Mathematics); Discrete Structures (Computer Science)
- Fall 2024: Discrete Structures (Computer Science)
- Fall 2023: General Mathematics; Mathematics Content-II (Education)
- Summer 2023: Numerical Methods (Mathematics); Precalculus (Computer Science)
- Spring 2023: College Algebra (Business Administration)
- Winter 2023: Business Calculus (Business Administration)
- Fall 2022: General Mathematics; Mathematics Content-II (Education)
- Spring 2022: Sindh Talent Hunt Program (Foundation Semester Mathematics)
- Fall 2019: National Talent Hunt Program (Foundation Semester Mathematics)

Lecturer in Mathematics (BPS-17)

Government Degree College, Ghotki

 Sep 2023 – Oct 2025

 Ghotki, Sindh, Pakistan

Taught mathematics to college students.

Peer Tutor (Mathematics)

Mathematics Resource Center, Sukkur IBA University

📅 Aug 2017 – May 2023

📍 Sukkur, Sindh, Pakistan

Tutored undergraduate students in Calculus, Linear Algebra, Differential Equations, and General Mathematics.

RESEARCH

M.S. Research Project / Preprint

Sukkur IBA University

📅 2022

Numerical Solution to the Riemann Problem for the Liquid–Gas Two-Phase Isentropic Flow Model

This work studied numerical solutions of a Riemann problem for a liquid–gas two-phase isentropic flow model. The governing nonlinear coupled PDE system (liquid and gas densities and liquid velocity) was solved on a 1D space–time grid using Python. Three schemes (Lax–Friedrichs, Lax–Wendroff, and FORCE) were implemented and compared. The FORCE scheme (TVD) reduced spurious oscillations and captured key flow features such as smooth profiles, shock waves, and rarefaction.

Link to [arXiv:2512.02071](https://arxiv.org/abs/2512.02071)

B.S. Research Project / Preprint

Sukkur IBA University

📅 2019

Mathematical Modeling and Simulation of HIV Dynamics

This work developed and analyzed within-host HIV models describing healthy $CD4^+$ T cells, infected $CD4^+$ T cells, and free virus. Treatment effects were incorporated through control parameters representing reverse transcriptase inhibitors and protease inhibitors. Equilibria were derived and local stability was studied via linearization and the Hartman–Grobman theorem. Numerical simulations were performed using the classical fourth-order Runge–Kutta method to illustrate viral load and cell-population dynamics under different therapy strengths and initiation times.

Link to [arXiv:2512.13171](https://arxiv.org/abs/2512.13171)

CONFERENCES AND WORKSHOPS

4th International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)

📅 Feb 27–28, 2023

📍 Sukkur IBA University

Learned about recent research trends in applied mathematics and scientific computing; attended technical talks on modeling and computational methods.

Workshop on “Digital Image Processing”

📅 Nov 22, 2022

📍 Sukkur IBA University

Gained hands-on exposure to core image processing concepts, filtering techniques, and basic implementation ideas for numerical experiments.

Workshop on “Scientific Computing on GPU: A Hands-on Approach”

📅 Jul 8–11, 2022

📍 Sukkur IBA University

Learned GPU-based scientific computing workflows and how parallel computation can accelerate numerical simulations.

3rd International Conference on Computing, Mathematics and Engineering Technologies

📅 Jan 29–30, 2020

📍 Sukkur IBA University

Strengthened understanding of interdisciplinary applications of mathematics; engaged with research presentations in computation and engineering.

2nd IEEE International Conference on Computing, Mathematics and Engineering Technologies (iCoMET 2019)

📅 Jan 30–31, 2019

📍 Sukkur IBA University

Observed how mathematical modeling and numerical methods support engineering and technology problems; learned about research communication through talks and poster sessions.

International Conference on Computing, Mathematics and Engineering Technologies (iCoMET 2018)

📅 Mar 3–4, 2018

📍 Sukkur IBA University

Developed early exposure to conference environments and emerging topics in applied mathematics, computing, and interdisciplinary research.

AWARDS AND HONORS

Awardee

Sindh Talent Hunt Program 2015 Scholarship

📅 Aug 3, 2015

📍 Sukkur IBA University

Monthly stipend, free accommodation, and full tuition fees

Awardee

Sindh Endowment Fund Scholarship

📅 Aug 3, 2015

📍 Sukkur IBA University

Full tuition fees waived for 4 years

Awardee

Prime Minister's Laptop

📅 Jul 2017

📍 Sukkur IBA University

Core i7 (8th Gen) Dell laptop and certificate of appreciation

Winner

Sports Gala Cricket Competition 2023

📅 Jan 15, 2023

📍 Sukkur IBA University

Trophy and winning certificate

SKILLS

LaTeX

Python

MATLAB

C++

MS Word

MS Excel

MS PowerPoint

Windows

Linux

COURSES OF INTEREST

Numerical Analysis

Numerical Methods

Numerical Methods for PDEs

Numerical Solutions of SDEs

Measure Theory

Real Analysis

Number Theory

Discrete Structures

Stochastic Calculus

PERSONAL INTERESTS

Cricket

Reading News Articles

Swimming

Gym

Research

Conferences

Movies

Adventure